

The big picture

Source-backed arXiv sample

Abstract

We discuss the conformal field theory and string field theory of the NSR superstring using a BRST operator with a nonminimal term, which allows all bosonic ghost modes to be paired into creation and annihilation operators. Vertex operators for the Neveu-Schwarz and Ramond sectors have the same ghost number, as do string fields. The kinetic and interaction terms are the same for Neveu-Schwarz as for Ramond string fields, so spacetime supersymmetry is closer to being manifest. The kinetic terms and supersymmetry don't mix levels, simplifying component analysis and gauge fixing.

1 Source Metadata

This sample is based on arXiv record `hep-th/9108021`. Authors: N. Berkovits, M.T. Hatsuda, and W. Siegel.

2 Abstract Discussion

We discuss the conformal field theory and string field theory of the NSR superstring using a BRST operator with a nonminimal term, which allows all bosonic ghost modes to be paired into creation and annihilation operators. Vertex operators for the Neveu-Schwarz and Ramond sectors have the same ghost number, as do string fields. The kinetic and interaction terms are the same for Neveu-Schwarz as for Ramond string fields, so spacetime supersymmetry is closer to being manifest. The kinetic terms and supersymmetry don't mix levels, simplifying component analysis and gauge fixing.